Re-analysis Summary Report for Peer-evaluation

Paper ID: **Anderson_AmEcoJourn_2011_bLe8**

Re-analyst's ID: **t89fm**

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: https://forms.gle/EAREaC6JjN2PWFfR6

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:
… [the study] finds substantially higher income for low-caste households residing in villages dominated by BACs [compared to villages dominated by upper caste]. (p. 240.)

The corresponding article

Here you can find the original article: https://osf.io/hcxdq/files/osfstorage/5e56aac535d22d02f75f482c?view_only=32ccf80d2c684ea9ac10ebbc4ec34df6
and the original data: https://osf.io/k7xzy/?view_only=c1f2b9aaf05d486c942783efd8ec57ec

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:
In my analysis I use multiple statistical methods to test whether low-caste households residing in villages dominated by BACs have significantly higher income than the ones residing in villages dominated by upper caste. As a first step I inspect the dependent variable's most important descriptive statistics and distribution. The total income of households in the sample is heavily right skewed as there are several households with extremely high income. Furthermore, the households with the highest total income are almost all from villages dominated by BACs. In order to handle these outlying values and avoid any potential bias, I use the logarithmic values of total income in the followings. Average log total income in households residing in villages dominated by BACs is 8.01, while in the other villages it is 7.68. Applying one-sided t-test to the means of the two groups, the null hypothesis that the two means are equal can be rejected with p-value close to zero. Consequently, the average total income in BACs dominated villages is significantly higher than in high caste dominated ones. However, the t-test does not control for any explanatory variables. Therefore, I build a linear regression model including all available variables as explanatory variables. The original hypothesis is tested via a dummy variable representing the dominant group of the village (domlow). The

explanatory power of the model is moderate with adjusted R2 equal to 56.1%, but the p-value of the partial t-test of the domlow variable with robust standard errors is higher than 5% (5.4%), thus it is considered insignificant. Executing backward elimination, I sequentially omit the variables with the least explanatory power until the Akaike information criteria of the model improves. The adjusted R2 of the restricted model is also 56.1% but the p-value of the domlow variable increased to 7.5%. In conclusion, the linear regression model results show that controlling for multiple variables, the total income in BACs dominated villages is not higher significantly. Finally, I apply Blinder-Oaxaca decomposition to the data. The Blinder-Oaxaca method decomposes differences in mean outcomes across two groups into explained and unexplained parts. The explained part is due to group differences in the levels of explanatory variables, while the unexplained is due to differential magnitudes of regression coefficients. There are multiple ways to choose the reference coefficients for the decomposition. In my analysis I use the average coefficients of the regression on the two groups with equal weights as proposed by Reimers (1983). Blinder-Oaxaca decomposition method has been widely applied in economic research, mainly to study labour market discrimination. The greater the unexplained part of the wage gap, the more likely that there is discrimination in the market. After removing the observations with missing data, the log total income in BACs and high caste dominated villages are 7.95 and 7.72, respectively. Thus, the aim of the Blinder-Oaxaca method is to decompose the 0.225 difference between the averages. The twofold decomposition shows that 0.209 log total income difference is due to group differences in the levels of explanatory variables (explained part) and only 0.016 difference is due to the difference in the regression coefficients. In addition, the bootstrapped standard errors (0.073 and 0.071 for explained and unexplained parts, respectively) suggest that the unexplained component does not significantly differ from 0. Therefore, total income difference between BACs and high caste dominated villages can be almost entirely explained by group differences in the levels of explanatory variables. To conclude, controlling for the right skewed distribution of the dependent variable and for multiple other socioeconomic factors, I found that low-caste households residing in villages dominated by BACs do not have significantly higher income than the ones residing in villages dominated by upper caste.

The analyst's conclusion: Low-caste households residing in villages dominated by BACs do not have significantly higher income than the ones residing in villages dominated by upper caste.

The analyst's categorization of the result:
The results show evidence for opposite relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:
You should not use District controls, Crop controls, Distance controls, Groundwater controls, Public goods controls Do not include Irrigation, credit, and tenancy variables, and their interaction term versions Do not use instruments for water buyer status.

Here, the analyst reported the following steps and results of the analysis:
In my analysis I use linear regression to test whether low-caste households residing in villages dominated by BACs have significantly higher income than the ones residing in villages dominated by upper caste. As a first step I inspect the dependent variable's most important descriptive statistics and distribution. The total income of households in the sample is heavily right skewed as there are several households with extremely high income. Furthermore, the households with the highest total income are almost all from villages dominated by BACs. In order to handle these outlying values and avoid any potential bias, I use the logarithmic values of total income in the followings. Then, I build a linear regression model including all variables that are not excluded from the analysis as explanatory variables. The original hypothesis is tested via a dummy variable representing the dominant group of the village (domlow). The explanatory power of the model is moderate with adjusted R2 equal to 45.8%, but the p-value of the partial t-test of the domlow variable with robust standard errors is higher than 5% (5.6%), thus it is considered insignificant. Since all the variables included in the analysis are significant except for the domlow no further model selection is necessary. To conclude, controlling for the right skewed distribution of the dependent variable and for other socioeconomic factors, I found that low-caste households residing in villages dominated by BACs do not have significantly higher income than the ones residing in villages dominated by upper caste.

Analysis language/software/code (optional to check):
Task 1: R
Task 2: R
Analysis codes: https://osf.io/wnr6u/